Weilin Sean Wan

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Education

Duke University & Duke Kunshan University **August 2023–Present (Expected May 2027)**
Dual-Degree Undergraduate Program *Durham, North Carolina, U.S.A.& Kunshan, Jiangsu, China*
Duke University: *B.S. in Interdisciplinary Studies*
Duke Kunshan University: *B.S. in Applied Mathematics and Computational Sciences, Computer Science Track*
Duke University Exchange Student: August 2025–December 2025, Durham, North Carolina, U.S.A.
Cumulative GPA: 3.810/4.000
Relevant Coursework: Data Acquisition and Visualization (A+); Computer Organization and Programming (A+); Artificial Intelligence (A); Principles of Machine Learning (A); Introduction to Databases (A); Discrete Mathematics for Computer Science (A); Multivariable Calculus (A-); Design and Analysis of Algorithms (A-); Probability and Statistics (A-); Introduction to Data Science (A-)

Research Interests

Multimodal AI, Vision Language Models and Reasoning.

Publications and Manuscripts

Peer-Reviewed and Accepted Papers

[1] **Sean Wan**, Shilin Ou, and Luyao Zhang. "ViRel-Budget: Reliability-Constrained Visual-Token Budgeting for Green Vision-Language Inference." ***GreenMM 2026 at ACM Multimedia 2026*****.** Accepted for Publication.

[2] **Sean Wan**, Dongping Liu, and Luyao Zhang. "StableEval Arena: A Cost-Aware Agentic Benchmark for Stablecoin Price Stability Prediction." ***KDD 2026 Workshop on Evaluation and Trustworthiness of Agentic AI*****. Selected Oral Presentation.**

[3] **Sean Wan**, Jiabao Brad Wang, and Mustafa Mısır. "Active Learning for Molecular-Docking Algorithm Selection under Heterogeneous Runtime." ***NeurIPS 2026 Workshops on AI for Drug Discovery: Bridging the Translation Gap*****.**

Manuscripts in Preparation

[4] **Sean Wan**, Kaizhu Huang. "StateDelta: State-Transition Grounded Instruction Generation from Procedural Image Sequences." To be submitted to ***ACL Rolling Review (ARR), January 2027 Cycle.*****.**

[5] **Sean Wan**, Haoyuan Zheng, Henghao Jiang, Zhaoxu Lin, Siyong Luo, and Mustafa Mısır. "Generative-AI-Assisted Portfolio Algorithm Selection under Asset-Family Shift." To be submitted to ***AAAI 2027 Workshop on Foundation Models and Generative AI for Finance (FinFM)*****.**

Research Experience

StateDelta: Grounded Multimodal Learning and Reasoning **June 2026–Present**
Undergraduate Signature Work. Advisor: Kaizhu Huang *Duke Kunshan University*

- Developing a **grounded assembly dataset** linking manual diagrams to detailed instructions, component annotations, and source evidence through model-assisted labeling and comprehensive human review, providing reusable supervision for retrieval evaluation and VLM fine-tuning.
- Designing a **RAG–VLM framework**: structured visual evidence → semantic and part-based retrieval → image-conditioned instruction generation, with subsequent VLM fine-tuning and RAG integration aimed at transferring knowledge from a curated assembly corpus across furniture designs and categories, while grounding instructions in specific components, attachment locations, and supported operations.
- Planned submission: ACL Rolling Review (ARR), January 2027 Cycle, targeting ACL 2027.

ViRel-Budget: Reliable and Efficient Vision-Language Inference **May 2026–August 2026**
Advisor: Luyao Zhang *Duke Kunshan University*

- Developed ViRel-Budget, a **reliability-constrained pipeline** for visual-token budgeting: construct safety labels measuring preservation of dense-model responses on original images and eligible visual interventions, train budget-specific risk predictors using pre-inference features, and calibrate a policy that selects a pruning budget or dense fallback for a single VLM call without backbone fine-tuning.
- Benchmarked on POPE, MMStar, and Visual CounterFact through group-isolated prospective evaluation and independent replication draws with paired GPU telemetry. The SCOPE-based controller achieved **replicated energy savings** of 11.87% and 10.62% versus matched dense LLaVA-1.5-7B and -13B inference, respectively, while meeting the calibrated unsafe-acceptance limit.
- **First-authored** paper accepted for publication at the **ACM Multimedia 2026 GreenMM** workshop.

StableEval Arena: Reliable Evaluation of Agentic AI Systems **April 2026–July 2026**
Advisors: Luyao Zhang and Dongping Liu *Duke Kunshan University*

- **Built StableEval Arena**, an environment-grounded historical-replay benchmark for agentic stablecoin risk assessment, with leakage-safe evidence windows, hidden future horizons, structured predictions, deterministic labeling, calibration, and reproducible cost/latency logging.
- **Benchmarked six agents** against always-stable, historical-persistence, threshold-rule, and logistic-regression baselines across stress-enriched and natural-distribution settings; calibrated agents surpassed historical persistence on aggregate performance in the enriched evaluation while revealing substantial rare-event failures overlooked by accuracy alone.
- **First-authored** paper accepted with **selected oral presentation** at the **KDD 2026 Workshop on Agentic AI Evaluation and Trustworthiness**.

Molecular Docking: Cost-Efficient Learning for Algorithm Selection **March 2026–August 2026**
DKU Fully Funded Summer Research Scholar. Advisor: Mustafa Mısır *Duke Kunshan University*

- Built a retrospective **active-learning pipeline** for molecular-docking algorithm selection, integrating cell-wise acquisition, low-rank matrix completion, and runtime models fitted only to previously observed training-pool runtimes; trained downstream selectors on completed performance matrices while reserving held-out docking outcomes for evaluation.
- Benchmarked **runtime-aware acquisition** against Mean-UCB through realized docking cost, per-engine query allocation, and held-out selector performance; demonstrated that partial acquisition yields decision-useful training data, identified budget-dependent cost–quality trade-offs, and showed that matrix-reconstruction metrics do not consistently track downstream docking performance.
- **First-author** paper accepted at the **NeurIPS 2026 Workshop on AI for Drug Discovery (AI4DD)**.

Research Experience Continued

Generalizable Algorithm Selection under Distribution Shift **July 2025–March 2026**
DKU & SUFE Fully Funded Summer Research Scholar. Advisor: Mustafa Mısır *DKU and Shanghai University of Finance and Economics*

- Developed a **portfolio algorithm-selection** pipeline evaluating executable LLM-generated features, pretrained time-series embeddings, and portfolio-trained Transformer representations with RF/MLP selectors to choose among classical and learned portfolio optimizers.
- Investigated **representation transfer** through CKA, controlled fine-tuning, and family-held-out and forward-time evaluations; portfolio-trained features increased mean annualized return by 12.4 percentage points over a market-feature RF selector in the 2025 forward test.
- Planned submission: AAAI 2027 Workshop on Foundation Models and Generative AI for Finance (FinFM).

Teaching Assistant Experience

Teaching Assistant, DKU STATS 302 / COMPSCI 309 Principles of Machine Learning	Fall 2026
Teaching Assistant, DKU COMPSCI 203 Discrete Mathematics for Computer Science	Spring 2026
Teaching Assistant, DKU COMPSCI 201 Introduction to Programming and Data Structures	Spring 2026
Teaching Assistant, DKU MATH 201 Multivariable Calculus	Spring 2026
Peer Tutor, DKU STATS 302 / COMPSCI 309 Principles of Machine Learning	Fall 2026–Spring 2027
Peer Tutor, DKU GCHINA 101 China in the World	Spring 2025

Honors and Awards

Summer Research Scholar, Duke Kunshan University (**Fully Funded**)	April 2026
DKU–SUFE Research Scholar (**Fully Funded**)	June 2025
Summer Research Scholar, Duke Kunshan University (**Fully Funded**)	April 2025
National Encouragement Scholarship (China)	October 2026
Student Leader of the Year, Duke Kunshan University (Community Selected)	April 2026
Outstanding Peer Tutor Award, Duke Kunshan University (Community Selected)	May 2025
Dean's List with Distinction, Duke Kunshan University	Spring 2026
Dean's List, Duke Kunshan University	Fall 2025
Dean's List with Distinction, Duke Kunshan University	Fall 2024
Entrance Scholarship, Duke Kunshan University (25% tuition)	August 2023–May 2027

Selected Professional Experience

Liberata (Duke-Funded Startup) **July 2025–Present**
Telemetry Team Lead; Machine Learning Engineer *Remote & Durham, North Carolina*

- Extended a Python **multi-agent simulation** of peer-review markets with continuous and discrete effort models, reviewer-dependent pricing, and nonlinear review rewards; developed multi-seed experiment support and agent-level diagnostics to compare reinforcement-learning policies with heuristic and stochastic baselines and investigate how incentives shape effort allocation between research and peer review.
- Designed **behavioral telemetry** across nine platform workflows, defining 142 event types and measures of task completion, abandonment, latency, and interaction friction to distinguish usability problems from technical failures; implemented TypeScript tracking helpers, structured event validation, and sensitive-data filtering to support consistent, interpretable behavioral analysis.
- Led **telemetry engineering** from measurement specification through implementation and technical handoff, translating product requirements into frontend/backend instrumentation contracts, integration guidance, and staging QA criteria; coordinated integration and testing requirements with platform engineers, establishing consistent event definitions and privacy boundaries across authentication, manuscript submission, peer review, and replication workflows.

Leadership and Service

Co-Chair, DKU Student Leaders Board **January 2026–December 2026**

- Led advocacy on research access, summer housing, and academic integrity; established student-feedback mechanisms, represented DKU in inter-university exchanges, and coordinated logistics, scheduling, and feedback collection for campus events with 300+ participants.

Orientation Leader, Duke University and DKU **June–August 2024, 2025, 2026**

- Facilitated onboarding, campus tours, and community-building for 500+ incoming students across Duke University and DKU over three years; coordinated orientation logistics to support students' transition into university life.

Project Lead, U-Corp–PEER Sustainability Education Initiative **March 2024–September 2024**

- Led delivery of a sustainability education initiative for 153 rural middle-school students through DKU's Innovation and Entrepreneurship Office and PEER; presented project outcomes at the 2024 Duke International Forum.

Co-Founder, VP of Growth and Strategy, and Treasurer, DKU Consulting Club **August 2024–May 2025**

- Co-founded the club and led growth strategy and treasury operations; drove 40% membership growth, recruited 113 new members, and guided market-research and FinTech report projects.

Technical Skills and Languages

Programming and Tools: Python, Java, C; scikit-learn, Jupyter, GitHub, LaTeX.
AI and Machine Learning: VLM training and evaluation, multimodal learning, agentic AI workflow construction, reinforcement learning.
Languages: Mandarin Chinese (native); English (fluent, TOEFL iBT 118/120); Spanish (intermediate); Cantonese (intermediate).